

Aditya Singh

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Summary

Computer Science undergraduate specializing in Artificial Intelligence and Machine Learning with hands-on experience in building end-to-end ML pipelines. Skilled in data preprocessing, feature engineering, model training, and evaluation using Python and Scikit-learn. Strong foundation in data structures, problem solving, and applied machine learning workflows.

Experience

Machine Learning Intern, Atharvo | Remote/India January 2024

- Processed and analyzed multi-city accident datasets using structured data transformation techniques.
- Performed feature engineering and supported predictive modeling workflows.
- Applied Random Forest for severity prediction and analyzed feature importance.
- Built visualization dashboards to communicate analytical insights effectively.

Data Science Intern, Unified Mentor Private Limited | Remote/India June 2025 – July 2025

- Developed structured data pipelines for multi-city accident datasets including ingestion, cleaning, and transformation.
- Applied feature engineering and supported predictive modeling workflows.
- Used SQL-style data aggregation and grouping techniques to derive business insights.
- Built visualization dashboards to monitor key performance metrics and data trends.

Machine Learning intern, EDXcellence | Remote/India Jan 2026 – March 2026

- Learned and implemented Machine Learning models using Python.
- Worked on data cleaning, preprocessing, model training, and evaluation.
- Used Pandas, NumPy, Scikit-learn, and visualization tools for ML tasks.
- Gained practical understanding of the ML workflow from dataset to prediction.

Education

K.R. Mangalam University, Bachelor of Technology in Computer Science and Engineering Aug 2023 – May 2027
(Expected)

Specialization: Artificial Intelligence and Machine Learning

- **Coursework:** Machine Learning, Artificial Intelligence, DSA, Python programming

Certifications

Big Data Engineer – IBM Oct 2025

Foundations of AI, Data Science & Analytics – Samatrix Group Jan 2024

Data Analysis With Python – IBM Developer Skills Network Jan 2024

Projects

Citadel — AI-Based Public Safety Alert System

- **Tech Stack:** Python, PyTorch, OpenCV, FastAPI, TypeScript, SQLite
- Built a real-time accident detection system using YOLO and DETR on a large-scale dataset of 1000+ Caltrans CCTV feeds.
- Developed video processing pipeline for frame-by-frame analysis and reliable detection across varying lighting and traffic conditions.
- Integrated Twilio-based alerting system for high-severity incidents with low-latency notifications.
- Designed backend using FastAPI to handle concurrent streams and deployed interactive map-based visualization for incident tracking.
- **Github Repository**

Supermarket Sales Prediction System

- **Tech Stack:** Python, Pandas, Scikit-learn, XGBoost, Streamlit, Matplotlib
- Built an end-to-end data processing pipeline for structured retail sales datasets.
- Engineered and transformed raw transactional data into model-ready format using feature extraction techniques.
- Implemented regression modeling using XGBoost achieving R^2 score of 0.91.
- Designed backend data logic for KPI computation and structured metric tracking.
- Maintained clean directory structure and version control (Git) following best software engineering practices.
- **Github Repository**

Technologies & Skills

- **Programming Languages:** Python, SQL
- **Machine Learning:** Supervised Learning, Regression, Classification, Model Evaluation, Feature Engineering
- **Data Processing:** Data Cleaning, Data Transformation, EDA, Structured Dataset Handling
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib
- **Core CS:** Data Structures & Algorithms, OOP
- **Tools:** Git, GitHub, Jupyter Notebook, Streamlit